Logging Slow Queries With Active Support Notifications

With Ruby on Rails and without any extra gems, plugins, or database extensions, a wealth of query information can be added to the Rails application log. *Active Support Notifications*[1] can be used to monitor slow queries. (See: *Track slow DB queries without additional gems*[2])

To see this in action, read and understand the *Slow Query Subscriber* class added to Rideshare that uses Active Support.

The *Subscriber* listens for events formatted as sql.active_record that have query information. The duration of the query can be calculated using the start and finish times associated with the event. When the duration exceeds 1 second, the full SQL statement can be logged using the Rails logger.

Review the class in Rideshare or copy this:

ruby/slow_query_subscriber.rb

```
# Inspiration: https://twitter.com/kukicola/status/1578842934849724416
class SlowQuerySubscriber < ActiveSupport::Subscriber
  SECONDS_THRESHOLD = 1.0

  ActiveSupport::Notifications.subscribe('sql.active_record')
  do |name, start, finish, _, data|

    duration = finish - start

    if duration > SECONDS_THRESHOLD
      sql = data[:sql]
      Rails.logger.debug "[#{name}] #{duration} #{sql}"
    end
  end
end
```

Open bin/rails console to test this out. By running the query SELECT pg_sleep(1); from Active Record as a query that simulates taking 1 second or more, you'll see the instrumentation get triggered and the following log output.

```
irb(main):004:0> ActiveRecord::Base.connection.execute("SELECT pg_sleep(1)")
[sql.active_record] 1.005035 SELECT pg_sleep(1)
   (1005.2ms)  SELECT pg_sleep(1)
```

In this technique you've logged a slow query from the client application side. What about when you want to view slow queries from the server side?

1. https://guides.rubyonrails.org/active_support_instrumentation.html
2. https://twitter.com/kukicola/status/1578842934849724416/photo/1